UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta preliminare del 09/11/2017*

PROBLEM

We observe N samples of the following discrete-time real random process:

$$X[n] = A \cdot s[n] + W[n], \quad n = 0, 1, \cdots, N-1,$$

where A is the amplitude of the Signal of Interest (SOI) and it is modelled as a random parameter, in particular it is modelled as a zero mean Gaussian random variable with variance σ_A^2, $s[n]$ is the signal shape, perfectly known, its energy is given by $E_s = \sum_{n=0}^{N-1} s^2[n] = N$, $W[n]$ is additive white Gaussian noise (AWGN) with Power Spectral Density (PSD) $S_W\left(e^{j2\pi f}\right) = \sigma_W^2$, where σ_W^2 is a-priori known.

(1) Derive the joint probability density function (pdf) of the received data vector $\mathbf{X} \triangleq \left[X[0] \quad X[1] \quad \cdots \quad X[N-1]\right]^T$ and of the random parameter A (this pdf can be expressed either in scalar form or in vector form, provide both). Then, derive the marginal pdf of the received data vector $\mathbf{X}$.

(2) Under the assumption that σ_W^2 is a-priori known, derive the Maximum A-Posteriori (MAP) estimator of the amplitude A and the pdf of the estimation error $\varepsilon \triangleq A - \hat{A}_{MAP}$. Express these quantities in terms of the SNR $\gamma \triangleq \sigma_A^2 / \sigma_W^2$ and establish whether the estimator is biased and consistent.

(3) Derive the Bayesian Cramèr-Rao lower bound (CRB), compare it with the Mean Square Error (MSE) of the MAP estimator and establish whether the estimator is efficient.

Solution

(1) Let's define $X_n \triangleq X[n]$ and $s_n \triangleq s[n]$. The pdf of the observed data vector conditioned to A is given by:

$$X_n | A \in N\left(As_n, \sigma_W^2\right), \quad \left\{X_n | A\right\}_{n=0}^{N-1} \text{ conditionally independent } \quad \Rightarrow \quad \mathbf{X} | A \in N\left(A\mathbf{s}, \sigma_W^2 \mathbf{I}\right)$$

$$A \in N\left(0, \sigma_A^2\right)$$

The pdf in scalar format is:

$$\begin{aligned} f_{X,A}(\mathbf{x}, a) &= f_{X|A}(\mathbf{x}|a)\, f_A(a) = \prod_{n=0}^{N-1} f_{X|A}(x_n|a)\, f_A(a) \\ &= \left(\frac{1}{\sqrt{2\pi\sigma_W^2}}\right)^N \exp\left(-\frac{1}{2\sigma_W^2}\sum_{n=0}^{N-1}\left(x_n - a\cdot s_n\right)^2\right)\cdot\frac{1}{\sqrt{2\pi\sigma_A^2}}\exp\left(-\frac{a^2}{2\sigma_A^2}\right) \end{aligned}$$

In vector format:

$$\begin{aligned} f_{X,A}(\mathbf{x}, a) &= f_{X|A}(\mathbf{x}|a)\, f_A(a) \\ &= \frac{1}{\left(2\pi\sigma_W^2\right)^{N/2}} \exp\left(-\frac{1}{2\sigma_W^2}(\mathbf{x} - a\mathbf{s})^T(\mathbf{x} - a\mathbf{s})\right)\cdot\frac{1}{\sqrt{2\pi\sigma_A^2}}\exp\left(-\frac{a^2}{2\sigma_A^2}\right) \end{aligned}$$

Pdf of $\mathbf{X}$ in vector format: $\mathbf{X} = A\mathbf{s} + \mathbf{W}$, $\mathbf{X}$ is a Gaussian vector, since $\mathbf{A}$ and $\mathbf{W}$ are Gaussian distributed: $\mathbf{X} \in N\left(\boldsymbol{\eta}_X, \mathbf{C}_X\right)$

$$\boldsymbol{\eta}_X = E\{\mathbf{X}\} = E\{A\mathbf{s} + \mathbf{W}\} = E\{A\}\mathbf{s} + E\{\mathbf{W}\} = \mathbf{0}$$

$$\mathbf{C}_X = E\left\{\mathbf{X}\mathbf{X}^T\right\} = E\left\{A^2\right\}\mathbf{s}\mathbf{s}^T + E\left\{\mathbf{W}\mathbf{W}^T\right\} = \sigma_A^2\mathbf{s}\mathbf{s}^T + \sigma_W^2\mathbf{I}$$

$$f_X(\mathbf{x}) = \frac{1}{(2\pi)^{N/2}\sqrt{\left|\sigma_A^2\mathbf{ss}^T + \sigma_W^2\mathbf{I}\right|}} \exp\left(-\frac{1}{2}\mathbf{x}^T\left(\sigma_A^2\mathbf{ss}^T + \sigma_W^2\mathbf{I}\right)^{-1}\mathbf{x}\right)$$

$$= \frac{1}{\left(2\pi\sigma_W^2\right)^{N/2}\sqrt{\left|\gamma\mathbf{ss}^T + \mathbf{I}\right|}} \exp\left(-\frac{1}{2\sigma_W^2}\mathbf{x}^T\left(\gamma\mathbf{ss}^T + \mathbf{I}\right)^{-1}\mathbf{x}\right)$$

where $\gamma \triangleq \sigma_A^2 / \sigma_W^2$

The inverse of the covariance matrix can be derived by using the *Woodbury's identity*:

$$(\mathbf{I} + \gamma\mathbf{ss}^T)^{-1} = \mathbf{I} - \frac{\gamma}{1+\gamma\mathbf{ss}}\mathbf{ss}^T = \mathbf{I} - \frac{\gamma}{1+N\gamma}\mathbf{ss}^T$$

The determinant is given by: $\left|\gamma\mathbf{ss}^T + \mathbf{I}\right| = 1 + N\gamma$. Hence, the pdf of **X** is given by:

$$f_X(\mathbf{x}) = \frac{1}{\left(2\pi\sigma_W^2\right)^{N/2}\sqrt{1+N\gamma}} \exp\left(-\frac{1}{2\sigma_W^2}\mathbf{x}^T\left(\mathbf{I} - \frac{\gamma}{1+N\gamma}\mathbf{ss}^T\right)\mathbf{x}\right)$$

(2) MAP estimator *A*:

$$\hat{A}_{MAP} = \arg\max_a\left(\ln f_{X,A}(\mathbf{x}, a)\right) = \arg\max_a\left(\ln f_{X|A}(\mathbf{x}|a) + \ln f_A(a)\right)$$

$$\ln f_{X|A}(\mathbf{x}|a) + \ln f_A(a) = \kappa - \frac{1}{2\sigma_W^2}\sum_{n=0}^{N-1}\left(x_n - a\cdot s_n\right)^2 - \frac{a^2}{2\sigma_A^2}$$

$$\frac{d}{da}\left(\ln f_{X|A}(\mathbf{x}|a) + \ln f_A(a)\right) = \frac{1}{\sigma_W^2}\sum_{n=0}^{N-1}s_n\left(x_n - a\cdot s_n\right) - \frac{a}{\sigma_A^2} = \frac{1}{\sigma_W^2}\left[\sum_{n=0}^{N-1}s_n x_n - aN - \frac{1}{\gamma}a\right] = 0$$

$$\Rightarrow \quad \hat{A}_{MAP} = \frac{\gamma}{1+N\gamma}\sum_{n=0}^{N-1}s_n x_n = \frac{\alpha}{N}\sum_{n=0}^{N-1}s_n x_n = \frac{\alpha}{N}\mathbf{s}^T\mathbf{x}$$

where $\alpha \triangleq \dfrac{N\gamma}{1+N\gamma}$

Since the estimate is a linear function of the Gaussian distributed data vector **X**, the estimate is Gaussian distributed, and consequently the estimation error is Gaussian distributed, with mean and variance given by:

$$E\left\{\hat{A}_{MAP}\right\} = E\left\{\frac{\alpha}{N}\mathbf{s}^T\mathbf{x}\right\} = \frac{\alpha}{N}\mathbf{s}^T E\{\mathbf{x}\} = \frac{\alpha}{N}\mathbf{s}^T\boldsymbol{\eta}_X = 0 \quad \Rightarrow \quad E\left\{A-\hat{A}_{MAP}\right\} = E\{A\} - E\left\{\hat{A}_{MAP}\right\} = 0$$

Hence, the estimator is unbiased.

$$\begin{aligned} MSE\left\{\hat{A}_{MAP}\right\} &= E\left\{\left(A-\hat{A}_{MAP}\right)^2\right\} = E\left\{\left(A-\frac{\alpha}{N}\mathbf{s}^T\mathbf{x}\right)^2\right\} = E\left\{\left(A-\frac{\alpha A}{N}\mathbf{s}^T\mathbf{s}+\frac{\alpha}{N}\mathbf{s}^T\mathbf{W}\right)^2\right\} \\ &= E\left\{\left((1-\alpha)A+\frac{\alpha}{N}\mathbf{s}^T\mathbf{W}\right)^2\right\} = (1-\alpha)^2\sigma_A^2 + \frac{\alpha^2}{N^2}\mathbf{s}^T E\left\{\mathbf{W}\mathbf{W}^T\right\}\mathbf{s} = (1-\alpha)^2\sigma_A^2 + \frac{\alpha^2}{N}\sigma_W^2 \\ &= \frac{1}{\left(1+N\gamma\right)^2}\sigma_A^2 + \frac{N\gamma^2}{\left(1+N\gamma\right)^2}\sigma_W^2 = \frac{\sigma_W^2\gamma}{1+N\gamma} = \alpha\frac{\sigma_W^2}{N} \end{aligned}$$

$$\lim_{N\to\infty} MSE\left\{\hat{A}_{MAP}\right\} = \lim_{N\to\infty}\alpha\cdot\lim_{N\to\infty}\frac{\sigma_W^2}{N} = 1\cdot 0 = 0$$

Hence, the estimator is consistent.

$$\varepsilon = A-\hat{A}_{MAP} \in N\left(0, \alpha\frac{\sigma_W^2}{N}\right)$$

(3) BCRB on the estimate of A:

$$\frac{d^2\ln f_{X,A}(\mathbf{x},a)}{da^2} = \frac{d^2}{da^2}\left(\ln f_{X|A}(\mathbf{x}|a) + \ln f_A(a)\right) = \frac{1}{\sigma_W^2}\cdot\frac{d}{da}\left[\sum_{n=0}^{N-1}s_n x_n - aN - \frac{1}{\gamma}a\right] = -\frac{1+N\gamma}{\gamma\sigma_W^2}$$

$$BCRB(A) = \frac{1}{-E\left\{\dfrac{d^2\ln f_{X,A}(\mathbf{x},a)}{da^2}\right\}} = \frac{\gamma\sigma_W^2}{1+N\gamma} = \alpha\frac{\sigma_W^2}{N} = MSE\left\{\hat{A}_{MAP}\right\}$$

Hence, the estimator is efficent.